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Configuration System

▼ Relevant source files

configIni.php

movetoparent/heuristConfigIni.php

The Heurist Configuration System provides a hierarchical approach to manage settings across multiple Heurist instances on a single server. This document explains how configuration values are defined, where they are stored, and how the override mechanism works to allow both shared configuration across instances and instance-specific customization when needed.

For information about database structure, see [Database Structure](#).

Configuration File Hierarchy

Heurist employs a two-tier configuration system where a parent configuration file provides default values for all instances, while individual instance configuration files can override specific settings as needed.

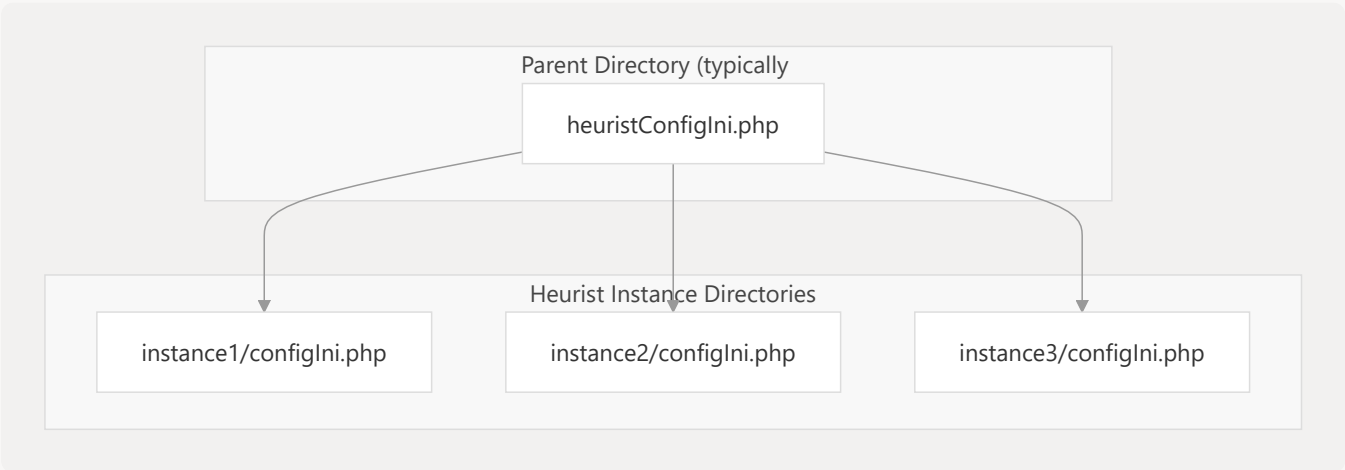


Figure 1: Configuration File Hierarchy

Sources: configIni.php 4-7 configIni.php 27-41 configIni.php 208-214

Configuration Files

Parent Configuration File

The parent configuration file (`heuristConfigIni.php`) resides in the parent directory of all Heurist instances, typically `/var/www/html/HEURIST/` . This file contains the shared configuration values that apply to all Heurist instances on the server, including:

- Database connection settings
- File paths and URLs
- Email addresses
- Administrator passwords
- Service integration settings (thumbnailing, SAML, etc.)
- Performance settings

During installation, a template file is moved from the codebase to the parent directory and renamed to `heuristConfigIni.php` .

Sources: `movetoparent/heuristConfigIni.php` 4-6 `configIni.php` 51-59

Instance Configuration File

Each Heurist instance has its own `configIni.php` file, which primarily contains:

1. The program version number (which should not be changed)
2. Default empty values for most configuration settings
3. Optional instance-specific overrides for any values that need to differ from the parent configuration

The instance configuration file is designed to be kept mostly empty, with values inherited from the parent configuration file.

Sources: `configIni.php` 3-6 `configIni.php` 63-72

Configuration Override Mechanism

The configuration system uses a simple but effective override mechanism:

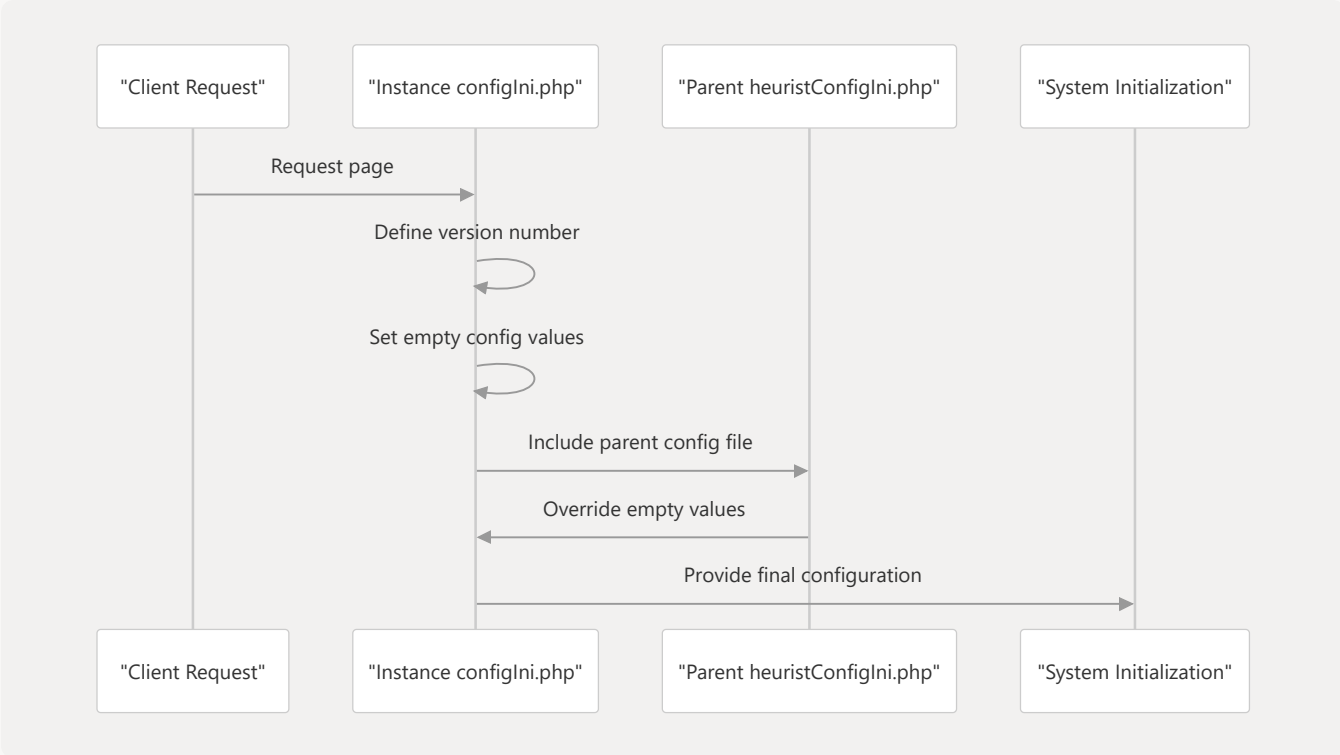


Figure 2: Configuration Loading Sequence

1. The instance's `configIni.php` is loaded first
2. It defines empty variables for configuration values
3. At the end, it includes the parent `heuristConfigIni.php` file
4. The parent file fills in the empty values with actual configuration
5. If the instance file has already set a value, it is not overridden by the parent file

This mechanism allows instance-specific configuration while minimizing duplication.

Sources: `configIni.php` 208-214 `movetoparent/heuristConfigIni.php` 35-36

Key Configuration Categories

The configuration system manages several categories of settings:

Category	Description	Examples
Database	Database connection settings	<code>\$dbHost</code> , <code>\$dbAdminUsername</code> , <code>\$dbPrefix</code>
Server	Server URLs and paths	<code>\$serverName</code> , <code>\$heuristBaseURL</code>
File Storage	File upload paths and URLs	<code>\$defaultRootFileUploadPath</code> , <code>\$defaultRootFileUploadURL</code>
Email	Email addresses for system functions	<code>\$sysAdminEmail</code> , <code>\$infoEmail</code> , <code>\$bugEmail</code>

Category	Description	Examples
Access Control	Administrator passwords	<code>\$passwordForDatabaseCreation</code> , <code>\$passwordForServerFunctions</code>
External Services	Integration with third-party services	<code>\$websiteThumbnailService</code> , <code>\$indexServerAddress</code>
Performance	Settings affecting system performance	<code>\$allowWebAccessThumbnails</code> , <code>\$useRewriteRulesForRecordLink</code>
Security	Security-related settings	<code>\$saml_service_provides</code> , <code>\$httpProxy</code>

Sources: `configIni.php` 85-193 `movetoparent/heuristConfigIni.php` 38-215

Configuration Data Flow

The following diagram illustrates how configuration data flows through the system:

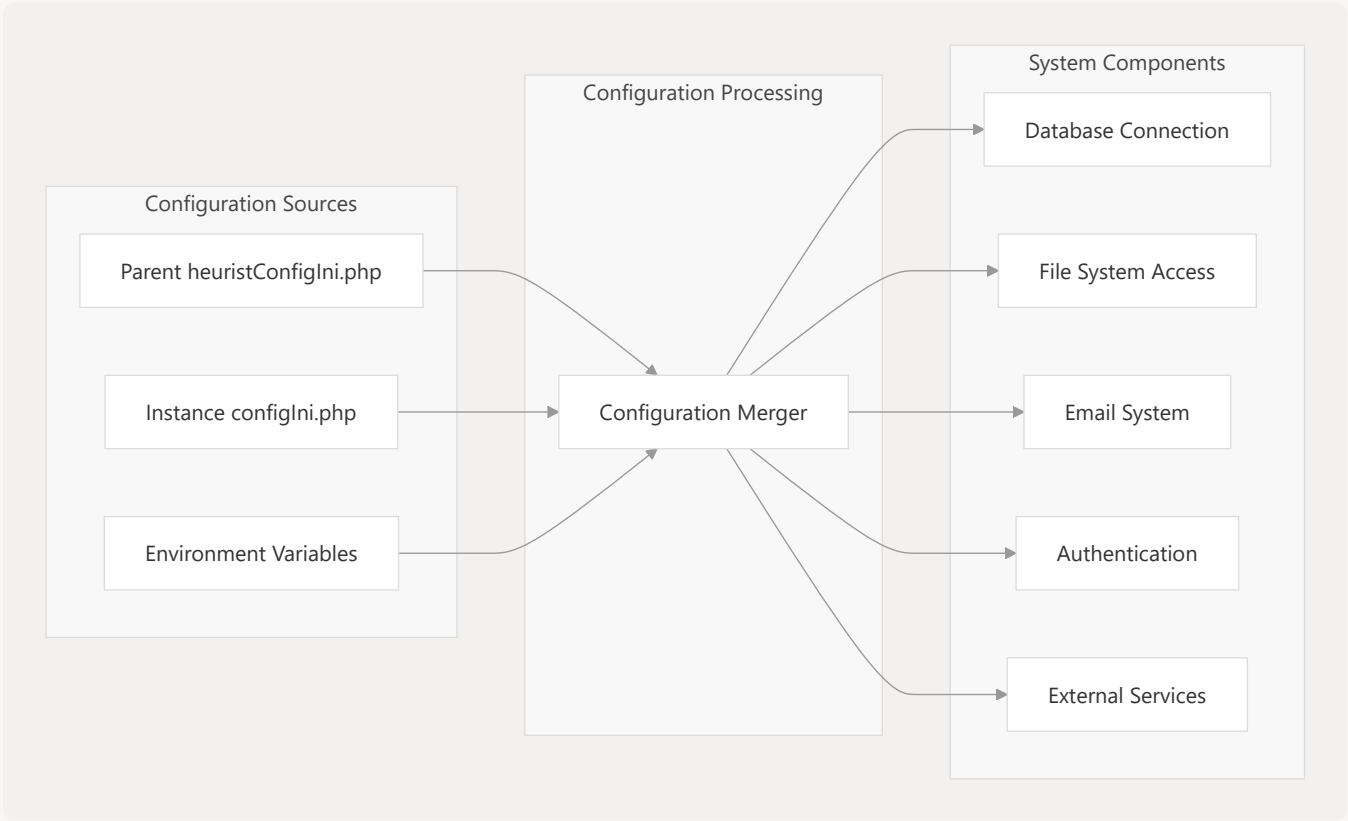


Figure 3: Configuration Data Flow

Sources: `configIni.php` 208-214 `movetoparent/heuristConfigIni.php` 60-66

Special Configuration Considerations

Version Number

The version number is always specified in the instance's `configIni.php` and should not be changed by users. It is not overridden by the parent configuration file.

```
$version = "6.7.2"; // sets current program version number, determined by Heurist development L
```

Sources: `configIni.php` 83-85

Reference Server

The Heurist reference server URL is critical for various functions and should not be changed:

```
$heuristReferenceServer = "https://heuristref.net"; // DO NOT CHANGE THIS
```

Sources: `configIni.php` 102-104

Database Prefix

A database prefix is strongly recommended to distinguish Heurist databases on your MySQL server:

```
$dbPrefix = 'hdb_'; // Strongly recommended to retain this
```

Sources: `configIni.php` 127

Environment Variable Support

Some configuration values, particularly sensitive database credentials, can be loaded from environment variables:

```
if (!@$dbAdminUsername) {  
    $dbAdminUsername = getenv("DB_ADMIN_USERNAME") ? getenv("DB_ADMIN_USERNAME") : "";  
}
```

This provides an additional security layer by keeping credentials out of configuration files.

Sources: `movetoparent/heuristConfigIni.php` 66-67

Best Practices

1. **Keep Instance Configuration Minimal:** Avoid setting values in the instance `configIni.php` unless necessary for special cases.
2. **Use Parent Configuration:** Set all shared values in the parent `heuristConfigIni.php` file.
3. **Protect Sensitive Information:** Use environment variables for sensitive credentials when possible.
4. **Don't Change Version Number:** The version number in `configIni.php` should be managed by the Heurist development team.
5. **Maintain Database Prefix:** Keep the recommended `hdb_` prefix for database names to distinguish Heurist databases.

Sources: `configIni.php` 30-41 `configIni.php` 74-81

Configuration Implementation Details

The implementation is straightforward - at the end of each instance's `configIni.php`, the parent configuration file is included if it exists:

```
$parentIni = dirname(__FILE__)."/../heuristConfigIni.php";

// parent directory configuration file is optional, hence include not required
// heuristConfigIni.php in parent directory overrides empty values in current file
if (is_file($parentIni)){
    include_once $parentIni;
}
```

This simple mechanism allows the parent file to provide values for any configuration variables that